Generative AI Applications Analysis Report

A Variational Autoencoder for Traffic Sign Image Generation, Capstone Project 5 (Generative AI Applications)

Overview

This report presents a Variational Autoencoder (VAE) that generates synthetic images of traffic signs, implemented in PyTorch. The motivating use case, carried over from Project 4's traffic sign classifier, is data augmentation: a generative model that can produce plausible synthetic sign images could in principle supplement a classifier's training data for sign types that are rare in real photographs. A VAE was chosen over a GAN because its training objective is a single differentiable loss rather than an adversarial game between two networks, which behaves more predictably on the comparatively small dataset used here. The full implementation is in the accompanying notebook (`generative_model.ipynb`).

Dataset Description

Project 4 used GTSRB, a dataset of German signs that are already cropped to individual images. We use the Mapillary Traffic Sign Dataset (MTSD), full street-level photographs with bounding-box annotations for every visible sign, crowd-sourced globally by Mapillary and released publicly for its validation split (Ertler, Mislej, Ollmann, Porzi, Neuhold, & Kuang, 2020). The data was obtained from the `ThankGod/mapillary_traffic_sign_dataset` Hugging Face mirror.

Two shard files from that mirror were used, giving 1,154 full photographs. Individual sign patches were cropped out of each photograph using its bounding-box annotations: each box was padded to a square centered on the box, clipped to stay inside the photograph, and resized to 32x32 pixels. Boxes smaller than 32 pixels on their shorter side, and boxes with an extreme aspect ratio (more than 2.5x longer in one dimension), were dropped; a visual check during development showed the latter were almost always mislabeled boxes around poles or signage edges rather than full signs. This produced 1,721 sign-crop training images. MTSD's 400 possible sign categories are spread thin across this sample: most represented categories have only a handful of crops, so the VAE was trained unconditionally as a general model of sign appearance rather than a model conditioned on a specific sign class.

Model Design and Training Approach

The encoder has three convolutional layers (3→32→64→128 channels, stride 2) that downsample the 32x32 input to a 4x4x128 feature map, flattened and projected to two 32-dimensional vectors, the mean and log-variance of an approximate posterior over the latent space. The decoder mirrors this in reverse with three transposed-convolution layers, ending in a sigmoid activation. Sampling uses the reparameterization trick, `z = mu + sigma * epsilon` with epsilon drawn from a standard normal, so gradients can flow through the sampling step during training (Kingma & Welling, 2014). Batch normalization after every convolution and transposed convolution helped stabilize training given the comparatively small dataset.

The training objective is the negative evidence lower bound (ELBO): a pixel-wise mean-squared-error reconstruction term plus a KL-divergence term pulling the approximate posterior toward the standard normal prior, with the KL term scaled by a weight of 0.5 (the beta-VAE formulation; Higgins et al., 2017). A beta below 1 was chosen to favor sharper reconstructions on this small dataset, at some cost to how smoothly the latent space matches the prior. The model trained for 150 epochs with the Adam optimizer (learning rate 1e-3, batch size 64) on a 90/10 train/validation split of the 1,721 crops.

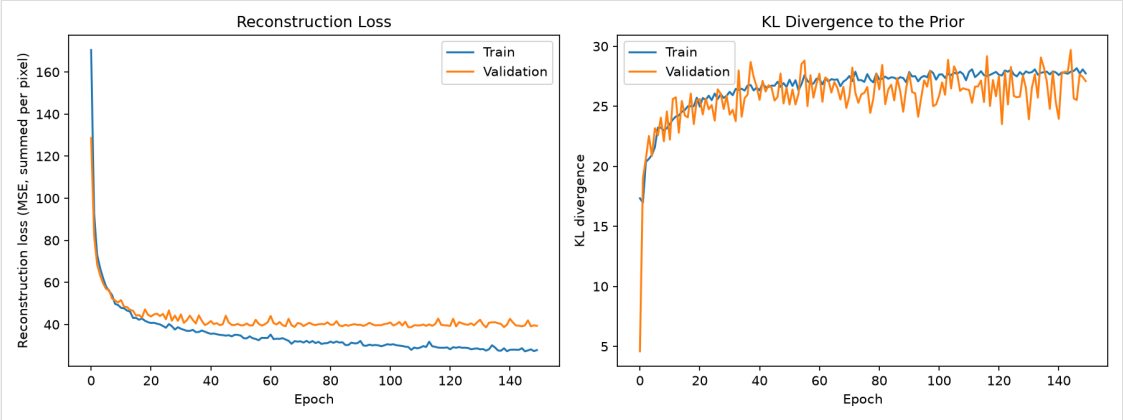


Figure 1. Training and validation reconstruction loss and KL divergence over 150 epochs.

Reconstruction loss dropped sharply over the first 20 epochs and then declined slowly, while the KL term rose over the same period before leveling off around epoch 40. Validation reconstruction loss settled about 10-12 points above training loss, and that gap stayed flat rather than widening over the remaining epochs, indicating a modest but stable generalization gap rather than progressive memorization of the training crops.

Output Evaluation and Interpretation

Before generating new images, the trained model's reconstructions of held-out validation crops were checked: the model preserves the overall shape and color of a real input sign (circle, diamond, triangle; red, blue, yellow) but loses fine detail such as printed numbers and pictograms, rendering them as a blurred smudge rather than a legible symbol.

New images were then generated by sampling latent vectors directly from the standard normal prior and decoding them, with no real image involved. Of 24 samples generated in the notebook, three (samples 15, 19, and 20) are the clearest successes: each shows a sharp red-and-white circular rim, the classic shape of a regulatory/prohibition sign, with a dark blob inside; sample 20 in particular shows two distinct dark marks side by side inside the rim, roughly where a two-digit speed-limit number would sit, though the marks are not legible as actual digits. The majority of the remaining samples are softer blue-toned circular blobs with a pale smudge in the middle and no visible rim. Two samples (18 and 22) are close to uniform dark smudges with no visible rim, border, or color contrast at all, an outright failure case with no interpretable sign structure. A different kind of failure appears in sample 5, which shows two overlapping circular shapes side by side rather than one sign, likely because some training crops included a second, partially visible sign at the edge of frame, which the model learned as a plausible pattern to reproduce. Across all 24 samples, nearly every one reads as a circle; no sample shows a clean diamond or triangle outline, even though both shapes are present in the training crops, suggesting the circular regulatory signs that are most common in this particular 1,721-crop sample dominate the latent space at the expense of rarer shapes.

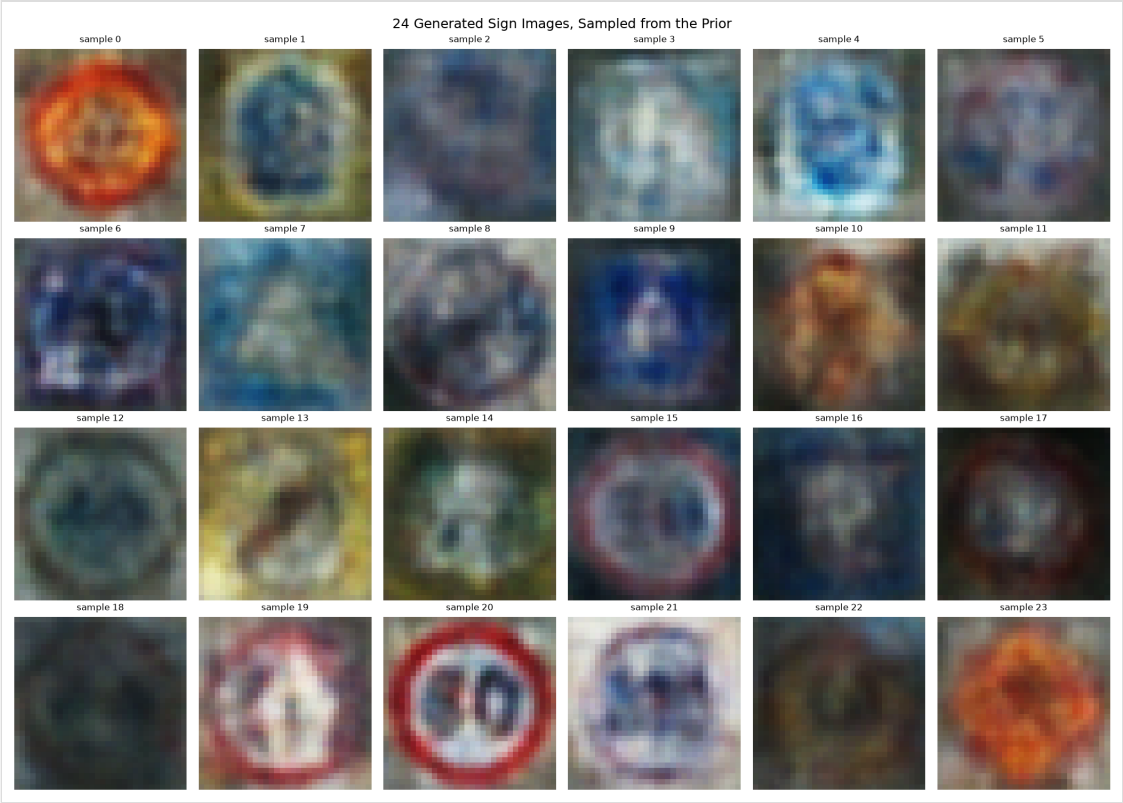


Figure 2. 24 images generated by sampling the prior and decoding, with no real image involved. Samples 15, 19, and 20 are the sharpest; samples 18 and 22 are failure cases with no visible sign structure.

Ethical Considerations and Responsible Use

The concrete failure modes observed above matter specifically because of the data-augmentation use case this notebook is framed around. If synthetic sign images like samples 18 or 22, which contain no genuine sign structure, or sample 5, which merges two signs into one implausible image, were added to a classifier's training set without screening, they would not represent any real sign class and could teach the classifier to associate a nonsense pattern with a label, or dilute its training signal with noise. Because the downstream use case in this capstone series is a traffic sign classifier intended as a component of an ADAS-style perception pipeline, this is not a cosmetic concern: a classifier trained partly on unrealistic synthetic data could develop subtle, hard-to-detect blind spots that only surface on specific real sign types, which is a safety-relevant failure mode, not a minor accuracy loss. Any real use of this kind of generator for augmentation would need a human or automated quality filter on generated images before they are added to a training set, and would need to report classifier performance separately with and without synthetic data so that this risk is visible rather than hidden inside an aggregate accuracy number.

A second consideration concerns the source data itself. MTSD's photographs were crowd-sourced from Mapillary contributors who captured and uploaded street-level imagery, and this project trains a model on crops of that imagery for a purpose (synthetic image generation) distinct from the original crowd-sourced mapping use. Responsible use of crowd-sourced street imagery for generative modeling should respect the dataset's stated license and research terms and should not be used to attempt to reconstruct or re-identify any people, vehicles, or private property that might incidentally appear in the source photographs.

Limitations and Future Improvements

No legible fine detail: as shown in both the reconstruction check and the generated samples, the model never renders a readable number, letter, or pictogram, only a blurred smudge, a known limitation of pixel-wise reconstruction loss in VAEs rather than a bug specific to this run.

Narrow shape diversity: generated samples are almost entirely circular, under-representing the diamond and triangle shapes present in training, likely because this project's two-shard sample of MTSD happens to contain more circular regulatory signs than warning signs.

Small, unconditional dataset: with only 1,721 crops spread across hundreds of sign categories, there is not enough data per category for class-conditional generation, so the model cannot be asked to generate, for example, a specific rare sign type on demand, which limits its direct usefulness for targeted augmentation of a classifier's weakest classes.

Future work could use more MTSD shards to get denser coverage per sign category, add class conditioning (for example a conditional VAE) so generation can be targeted at specific rare sign types, or replace the pixel-wise MSE loss with a perceptual or adversarial loss term to sharpen fine detail, since the current blurriness is the single biggest obstacle to using these outputs as real augmentation data.

References

Ertler, C., Mislej, J., Ollmann, T., Porzi, L., Neuhold, G., & Kuang, Y. (2020). The Mapillary traffic sign dataset for detection and classification on a global scale. In A. Vedaldi, H. Bischof, T. Brox, & J.-M. Frahm (Eds.), *Computer Vision – ECCV 2020* (Lecture Notes in Computer Science, Vol. 12368, pp. 68–84). Springer. https://doi.org/10.1007/978-3-030-58592-1_5

Higgins, I., Matthey, L., Pal, A., Burgess, C., Glorot, X., Botvinick, M., Mohamed, S., & Lerchner, A. (2017). beta-VAE: Learning basic visual concepts with a constrained variational framework. *Proceedings of the 5th International Conference on Learning Representations (ICLR 2017)*.

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